

Review – Exam 2

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Instruction: in a group of two, answer the following questions. Answer the questions without running the code in Visual Studio Code, or any other text editor (IDE).

Once all three exercises are completed, save this Word document as ***review_lastName***, and submit this word document individually, meaning that each student in a group must upload the answers to Blackboard.

Exercise 1) What is the output produced by the following program?

```
char mystery(int first, int second){ → first = -3 second = 1
    if (first >= second){           → true or false? false
        return 'W';                → return? 'W'
    }
    else{                           → run or skip run
        return 'H';                → return? 'H'
    }
}

int main()
{
    char m = mystery(-3, 1);
    cout<<"mystery returns = "<<m<<endl;
    return 0;
}
```

The print of the program should be: mystery returns = H

Exercise 2) Given the following code, complete the table below according to the **for** loop. **You can add more cells to the table if needed**

```
int sum=0, count=0;
for (int n = -10; n< 10; n+=3){
    if (n%4 ==0){
        count++;
        continue;
    }
    sum += n;
}
cout<<"The sum is: "<<sum<<endl<<"The count is: "<<count<<endl;
```

Loop iteration	n	if (n%4 ==0) True or False?	count	sum
1	-10	False	0	-10
2	-7	False	0	-17
3	-4	True	1	-17
4	-1	False	1	-18
5	2	False	1	-16
6	5	False	1	-11
7	8	True	2	-11
8	11 (Loops stop)	False	2	-11

What should be prompted at the end of the program?

```
cout<<"The sum is: "<<sum<<endl<<"The count is: "<<count<<endl;
```

The print of the program should be:

The sum is: -11

The count is: 2

Exercise 3) Given the following code, complete the table below according to the **while** loop. **You can add more cells to the table if needed**

```
int num1 = 23, num2 = 6, tensCount = 0, loopCount = 0;
while (num1 > 5 and num2<30){
    if(num1%2==1 or num2>10)
        tensCount ++;
    num1 -= 5;
    num2 += 5;
    loopCount ++;
}
cout<<"ten counter = "<<tensCount<<endl;
cout<<"loop counter = "<<loopCount<<endl;
```

Loop iteration	if (num1%2==1 or num2>10) True or False?	num1	num2	tensCount	loopCount
1	True	18	11	1	1
2	True	13	16	2	2
3	True	8	21	3	3
4	True	3	26	4	4
5	Loops stop	3	26	4	4
6	Loops Stop	3	26	4	4

What should be prompted at the end of the program?

```
cout<<"ten counter = "<<tensCount<<endl;
cout<<"loop counter = "<<loopCount<<endl;
```

The print of the program should be:

Ten counter = 4

Lopp counter = 4

Exercise 4) recursive function

Complete the following table according to the given code:

```
#include<iostream>
using namespace std;

// recursive sequence

int series(int m){
    if(m>=10)
        return m;

    else
        return series(m+2) - 6;
}

int main(){
    int sequencenumber = 3;
    cout<<"The series sequence of "<<sequencenumber<<" is " << series(sequencenumber)
    << endl;

    return 0;
}
```

Function iteration	int m	if(m>=10) true or false	else return series(m+2) - 6
1	3	False	$-7 - 6 = -13$
2	5	False	$-1 - 6 = -7$
3	7	False	$5 - 6 = -1$
4	9	False	$11 - 6 = 5$
5	11	True	11
6 (Loops stop)	skip	skip	skip

What should be prompted at the end of the program?

```
cout<<"The series sequence of "<<sequencenumber<<" is " << series(sequencenumber)
```

The print of the program should be:

The series sequence of 3 is -13
